

Pallavi Sinha

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SUMMARY

Full-stack engineer with 1+ year of production experience building distributed systems, real-time dashboards, and containerized infrastructure across a UAV-tech startup and a government organization. Delivered measurable impact: **84% API latency reduction** and **87% faster dev environment setup** in live systems. Proficient in Node.js, Angular 19+, Python, Redis, and Docker.

EXPERIENCE

- Software Trainee** Jan 2026 – Present
GarudaUAV Soft Solutions Pvt. Ltd. Noida, U.P.
- Built a **real-time S3 monitoring dashboard** with **Redis-backed job tracking** for 8 concurrent downloads; replaced polling with **event-driven architecture**, improving responsiveness and reducing network overhead.
 - Led architecture review of a **6-service distributed pipeline**, scoping 6 features across 4 cross-service dependencies within a 30-day timeline, unblocking cross-team delivery.
 - Authored technical documentation and served as engineering point-of-contact for stakeholder communication, supporting requirement clarification and iterative feature definition aligned to Scrum ceremonies.
- Software Engineering Intern** Jul 2024 – Aug 2024
National Informatics Centre CGO Complex, New Delhi
- Reduced **local environment setup time by ~87%** (15 min → under 2 min) by containerizing a multi-service dev stack with **Docker and Docker Compose**, enabling consistent, reproducible environments across the team.

PROJECTS

- Scalable Image Compression Service** | Node.js, Express, BullMQ, Redis, Sharp, MySQL, Cloudinary
- Designed and built a **distributed async job queue** using **BullMQ and Redis**, processing compression jobs across **multiple concurrent workers** with automatic retry logic and fault-tolerant failure handling.
 - Delivered **real-time job status via Server-Sent Events (SSE)**, eliminating client-side polling; persisted **job state and retry history in MySQL** for traceable, auditable processing.
- AI-Powered Medical Fact Checker** | Python, Qdrant, SPECTER2, Mistral API, RAG
- Built an end-to-end **RAG pipeline** indexing **150+ open-access research papers** in a **Qdrant** vector DB using **SPECTER2** biomedical embeddings, with top-k retrieval filtered at a **0.7 confidence threshold**.
 - Prompted **Mistral-Large** with source-attributed excerpts at **temperature 0.2** to produce grounded, citation-backed, low-hallucination responses across COVID-19, vaccines, and nutritional health domains.
- Blog Platform API** | Node.js, Express, MongoDB, Redis, JWT, BCrypt, Cloudinary
- Architected a **RESTful API** with **JWT authentication**, bcrypt password hashing, and **Redis session caching**, achieving sub-200ms latency across all authenticated routes.
 - Reduced **database query load by 60%** via Redis caching on high-traffic routes, cutting **average latency from 450ms to 70ms** — an **84% improvement** under simulated production traffic.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++, SQL, HTML, CSS
Frontend: Angular 19+, RxJS, React(portfolio, academic project), TailwindCSS, SASS
Backend: Node.js, Express, REST API design, JWT, OAuth2, Server-Sent Events (SSE)
Databases & Queues: MongoDB, MySQL, Redis, BullMQ, Qdrant, Mongoose
AI / ML: RAG pipelines, vector search, SPECTER2 embeddings, Mistral API, prompt engineering
DevOps & Cloud: Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Cloudinary
Tools: Postman, Git

ACHIEVEMENTS

- Smart India Hackathon — Winner** Dec 2023
Built **NyAI**, a legal documentation assistant simplifying legal jargon; led end-to-end UI/UX design and frontend implementation for the winning team.
- Code4Cause Hackathon — Runner Up** Oct 2023
Led a 6-member backend team; coordinated full-stack integration and system architecture under a 36-hour constraint.
- BNY Code Divas — DSA Round 1 Qualifier** Feb 2025
Selected among **92,000+** participants in a national competitive DSA assessment.

EDUCATION

Guru Gobind Singh Indraprastha University New Delhi
B.Tech, Electronics & Communication Engineering — Minor in ML & Data Analytics — CGPA: 8.64 / 10 2021 – 2025